have one item processed, which isn’t as good as with the division by data. However, once the pipeline is *primed*, things proceed a bit differently; after the first core has processed the first item, it moves on to the second, so once the final core has processed the first item, it can perform its step on the second. You now get one item processed every 3 seconds rather than having the items processed in batches of four every 12 seconds.

The overall time to process the entire batch takes longer because you have to wait 9 seconds before the final core starts processing the first item. But smoother, more regular processing can be beneficial in some circumstances. Consider, for example, a system for watching high-definition digital videos. In order for the video to be watchable, you typically need at least 25 frames per second and ideally more. Also, the viewer needs these to be evenly spaced to give the impression of continuous movement; an application that can decode 100 frames per second is still no use if it pauses for a second, then displays 100 frames, then pauses for another second, and displays another 100 frames. On the other hand, viewers are probably happy to accept a delay of a couple of seconds when they *start* watching a video. In this case, parallelizing using a pipeline that outputs frames at a nice steady rate is probably preferable.

Having looked at various techniques for dividing the work between threads, let’s take a look at the factors affecting the performance of a multithreaded system and how that can impact your choice of techniques.

## 8.2. Factors affecting the performance of concurrent code

If you’re using concurrency in order to improve the performance of your code on systems with multiple processors, you need to know what factors are going to affect the performance. Even if you’re just using multiple threads to separate concerns, you need to ensure that this doesn’t adversely affect the performance. Customers won’t thank you if your application runs *more slowly* on their shiny new 16-core machine than it did on their old single-core one.